# Uncovering biological relationships through morphological profiling: The JUMP-Cell Painting dataset and tools

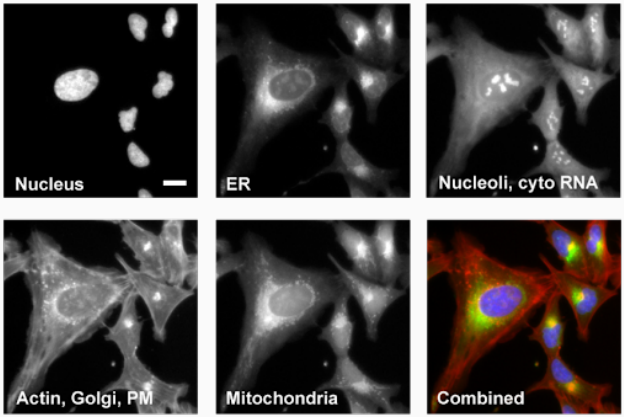
Alán F. Muñoz

Carpenter-Shen Lab, Purdue University

10 November 2026

# What is Cell Painting?

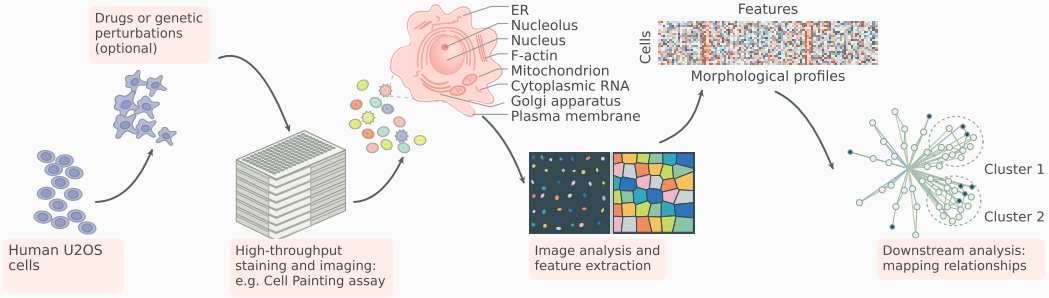
- Fluorescence microscopy of cell morphology.
- Six dyes, five imaging channels.
- Shape, intensity and texture reveal cellular responses.




Source: Muñoz, I2K (2024); assay: Bray et al. (2016)

# Cell Painting turns images into measurements

Profiles are numerical descriptions of cells.




Adapted from Chandrasekaran et al. (2021), Fig. 1a

# JUMP: Joint Undertaking for Morphological Profiling

- Public Cell Painting data: 12 laboratories, around 136,000 perturbations.
- 1.8 billion cells; ~58.3 million images.

| CRISPR knockout | Gene overexpression (ORF) | Compound treatment |
|---|---|---|
| **7,975** gene targets | **12,609** genes | **115,794** compounds |

# We make JUMP accessible via the JUMP Hub

## Choose Your Own Path

1. Just exploring the data?
   - Use a web interface; no programming required. You can look up images for a gene or compound, distinguishing features, most-similar genes or compounds, and more! See the Interactive Tools page for instructions for these tools.
2. Want to fetch and analyze data?
   - Follow our how-to guides for common analyses
3. Looking to build tools?
   - Use our Python libraries cp_measure, cpgdata, jump-portrait, jump-babel and other libraries in the monorepo
   - Access the metadata schema

## Biologists can find out:

- Does my gene or compound have a phenotype?
- Which perturbations cause similar phenotypes?
- What changes in cells and organelles?
- Is the phenotype heterogeneous across cells/samples?

Source: JUMP Hub (2026)

# Tutorials, how-tos and resources

The Hub contains practical guides to help biologists use JUMP.

| | |
|---|---|
| **How-to guides** | No-code gene/compound queries; explore similarity heatmaps. |
| **Tutorials** | Curated runnable notebooks for profiles, metadata and statistical analysis. |
| **Further resources** | Dataset explanations, protein-localisation analysis and a Guide to Cell Painting for Biologists. |

# A tale of two genes (that shape mitochondria)

**OPA1: mitochondrial fusion**

- Helps mitochondria fuse.
- Loss is expected to produce shorter, fragmented networks.

**DNM1L / DRP1: mitochondrial division**

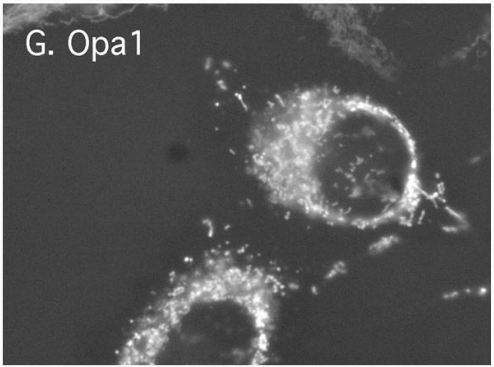
- Helps mitochondria divide.
- Loss is expected to produce longer, more-connected networks.

**We will compare knockouts and their negative controls.**

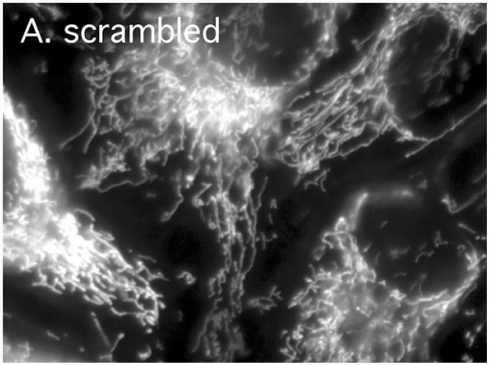
From Griparic et al., JCB (2007), Fig. 3; Smirnova et al., MBoC (2001).

# OPA1 depletion fragments mitochondria

**OPA1 siRNA**




**Scrambled siRNA**

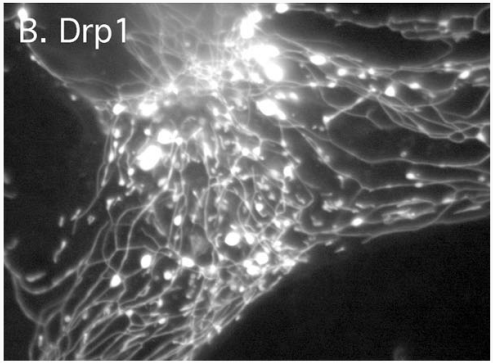



HeLa cells; MitoTracker. OPA1 depletion for 2 days.

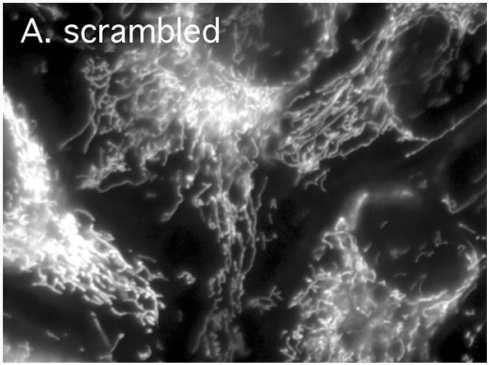
Griparic, Kanazawa & van der Bliek, JCB (2007), Fig. 3G/A. cropped

# DRP1 depletion increases connectivity

**DNM1L / DRP1 siRNA**




**Scrambled siRNA**




HeLa cells; MitoTracker. DRP1 depletion for 2 days.

Griparic, Kanazawa & van der Bliek, JCB (2007), Fig. 3B/A. Published microscopy, cropped.

# What do we want to find in JUMP?

**Find the perturbations for OPA1 and DNM1L.**

- Check which genetic perturbations are available.
- Identify the exact CRISPR reagent for each gene.
- Carry those identifiers into the image and feature tables.

**How:** search the gene-availability table, then filter JUMP_rr by gene and reagent.

JUMP Hub gene availability

# Can we see the expected phenotype in JUMP?

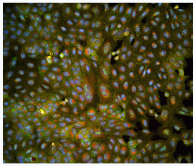
**1 row where Perturbation = "OPA1" and Plate = "CP-CC9-R1-06" sorted by rowid**

| | | |
|---|---|---|
| Perturbation | = | OPA1 |
| Plate | = | CP-CC9-R1-06 |
| - column - | = | |

Apply

✎ View and edit SQL

This data as json, CSV (advanced)

| Link | rowid ▼ | Perturbation | JCP2022 | Site 0 | Source | Plate | Well |
|---|---|---|---|---|---|---|---|
| 10049 | 10049 | OPA1 | JCP2022_804796 |  | source_13 | CP-CC9-R1-06 | D20 |

**Tip:** enlarge image previews on hover with Imagus (Firefox) or Hover Zoom+ (Chrome).

# Where can we see the perturbed cells?

**1 row where Perturbation = "OPA1" and Plate = "CP-CC9-R1-06" sorted by rowid**

Perturbation = OPA1

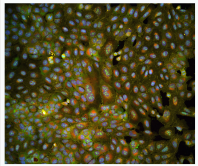
Plate = CP-CC9-R1-06

- column - =

Apply

✎ View and edit SQL

This data as json, CSV (advanced)

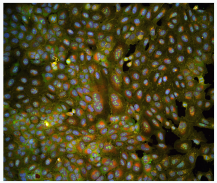
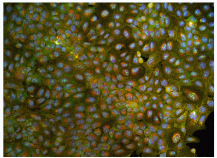
| Link | rowid ▼ | Perturbation | JCP2022 | Site 0 | Source | Plate | Well |
|---|---|---|---|---|---|---|---|
| 10049 | 10049 | OPA1 | JCP2022_804796 |  | source_13 | CP-CC9-R1-06 | D20 |

**How:** filter the image table for OPA1, then record source_13, plate CP-CC9-R1-06 and treatment well D20 before selecting controls.

# What should we compare those cells against?

This data as json, CSV (advanced)

Suggested facets: Perturbation, Resources, JCP2022

| Link | rowid | Perturbation ▼ | JCP2022 | Site 0 | Source | Plate | Well |
|---|---|---|---|---|---|---|---|
| 10049 | 10049 | OPA1 | JCP2022_804796 |  | source_13 | CP-CC9-R1-06 | D20 |
| 155 | 155 | no-guide | JCP2022_800001 |  | source_13 | CP-CC9-R1-06 | K02 |

**How:** retain source_13 and CP-CC9-R1-06, then add no-guide and non-targeting controls from that same plate.

# Is the gene represented in JUMP?

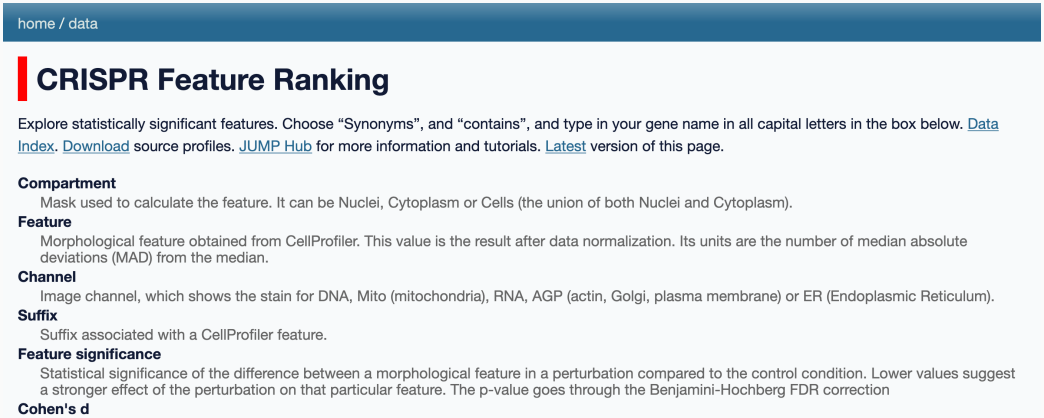
10 entries per page  Search: OPA

| Gene | CRISPR | ORF | A549 | HeLa_DMEM | H |
|---|---|---|---|---|---|
| str | i64 | i64 | i64 | i64 | |
| COPA | 0 | 1 | 0 | 0 | |
| OPA1 | 1 | 1 | 1 | 0 | |
| OPA3 | 1 | 1 | 0 | 0 | |
| OPALIN | 0 | 1 | 0 | 0 | |

Showing 1 to 4 of 4 entries (filtered from 15,696 total entries)  « ‹ 1 › »

**How:** search OPA, identify OPA1 among the returned genes, and check its CRISPR and ORF coverage before opening the other tables.

# Which table columns answer our question?



home / data

## CRISPR Feature Ranking

Explore statistically significant features. Choose “Synonyms”, and “contains”, and type in your gene name in all capital letters in the box below. Data Index. Download source profiles. JUMP Hub for more information and tutorials. Latest version of this page.

**Compartment**
Mask used to calculate the feature. It can be Nuclei, Cytoplasm or Cells (the union of both Nuclei and Cytoplasm).

**Feature**
Morphological feature obtained from CellProfiler. This value is the result after data normalization. Its units are the number of median absolute deviations (MAD) from the median.

**Channel**
Image channel, which shows the stain for DNA, Mito (mitochondria), RNA, AGP (actin, Golgi, plasma membrane) or ER (Endoplasmic Reticulum).

**Suffix**
Suffix associated with a CellProfiler feature.

**Feature significance**
Statistical significance of the difference between a morphological feature in a perturbation compared to the control condition. Lower values suggest a stronger effect of the perturbation on that particular feature. The p-value goes through the Benjamini-Hochberg FDR correction

**Cohen's d**

**How:** read the OPA1 row’s reagent, compartment, feature, channel, per-feature significance and signed effect size.

# How do we isolate the gene we care about?

**31 rows where Perturbation = "OPA1" sorted by Feature Rank**

Perturbation = OPA1

- column - =

Apply

✎ View and edit SQL

This data as json, CSV (advanced)

| Link | rowid | Compartment | Feature | Channel | Suffix | Feature significance | Cohen's d | \|Cohen's d\| | I |
|---|---|---|---|---|---|---|---|---|---|
| 243374 | 243374 | Cytoplasm | Texture_Correlation | Mito | _3_00_256 | 0.0 | 3.56 | 3.56 | |

**How:** filter by the exact symbol OPA1 or a published synonym such as MGM1, then add object-level filters when needed.

# Does the perturbation have a measurable phenotype?

**31 rows where Perturbation = "OPA1" sorted by Feature Rank**

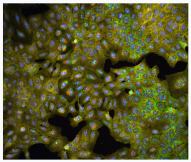
Perturbation = OPA1

- column - =

Apply

✎ View and edit SQL

This data as json, CSV (advanced)

| | Perturbation | Corrected p-value | Phenotypic activity | Perturbation example image | Median | Perturbation Rank | Feature Rank ▼ | JCP2022 |
|---|---|---|---|---|---|---|---|---|
| | OPA1 | 0.00018 | 0.85692 |  | 2.689 | 99999 | 0 | JCP2022_80 |

**How:** read the perturbation-level values: OPA1 CRISPR $p = 0.00018$, activity $0.85692$; OPA1 ORF $p = 0.03371$, activity $0.70875$.

# What does one feature result describe?

This data as json, CSV (advanced)

| Link | rowid | Compartment | Feature | Channel | Suffix | Feature significance | Cohen's d | \|Cohen's d\| | I |
|---|---|---|---|---|---|---|---|---|---|
| **243374** | 243374 | Cytoplasm | Texture_Correlation | Mito | _3_00_256 | 0.0 | 3.56 | 3.56 | |
| **63369** | 63369 | Nuclei | AreaShapeMaxFeretDiameter | | | 0.0 | -3.196 | 3.196 | |

**How:** read across one OPA1 row: reagent, object, feature, channel, significance, effect size and rank. It summarizes a profile, not one cell.

# Which features explain the differences?

**31 rows where Perturbation = "OPA1" sorted by Feature Rank**

Perturbation = OPA1

- column - =

Apply

✎ View and edit SQL

This data as json, CSV (advanced)

| Link | rowid | Compartment | Feature | Channel | Suffix | Feature significance | Cohen's d | \|Cohen's d\| | I |
|---|---|---|---|---|---|---|---|---|---|
| 243374 | 243374 | Cytoplasm | Texture_Correlation | Mito | _3_00_256 | 0.0 | 3.56 | 3.56 | |

**How:** sort OPA1 CRISPR rows by feature rank and compare significance with signed effect size; the leading row is a Mito texture feature.

# CellProfiler detects mitochondrial feature changes

| Gene | Feature | Cohen's $d$ |
|---|---|---|
| OPA1 | Mito texture correlation | $+3.56$ |
| DNM1L | Mito-skeleton branch ends | $-5.95$ |
| DNM1L | Mito granularity | $+4.09$ |
| DNM1L | Mito-skeleton total length | $-3.55$ |

**Mitochondrial organization changes.**

# What happens with a different perturbation type?

**30 rows where Perturbation = "OPA1" sorted by Feature Rank**

Perturbation = OPA1

- column - =

Apply

✎ View and edit SQL

This data as json, CSV (advanced)

| Link | rowid | Compartment | Feature | Channel | Suffix | Feature significance | Cohen's d | \|Cohen's d\| | Pe |
|---|---|---|---|---|---|---|---|---|---|
| **251862** | 251862 | Nuclei | Granularity_4 | Mito | | 0.03527 | 1.406 | 1.406 | OI |

**How:** open OPA1 in the ORF table and compare its separate reagent, activity and ranked feature effects with CRISPR.

# Which other perturbations have similar profiles?

**50 rows where Perturbation = "OPA1" sorted by Perturbation-Match Similarity descending**

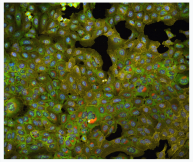
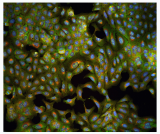
Perturbation = OPA1

- column - =

Apply

✎ View and edit SQL

This data as json, CSV (advanced)

| Link | rowid | Perturbation | Match | Perturbation example image | Match example image | Perturbation-Match Similarity ▲ | Match resources |
|---|---|---|---|---|---|---|---|
| 116095 | 116095 | OPA1 | MFN1 |  |  | 0.424 | NCBI, OMIM, GeneCards, Ensembl |

**How:** filter the CRISPR match table for OPA1 and sort similarity descending; MFN1 is the strongest published match shown ($0.424$).

# How do we keep and share the result?

**27 rows where Compartment in ["Cytoplasm", "Nuclei"] and Perturbation = "OPA1" sorted by Feature Rank**

Compartment | in | Cytoplasm,Nuclei

Perturbation | = | OPA1

- column - | = |

Apply

✎ View and edit SQL

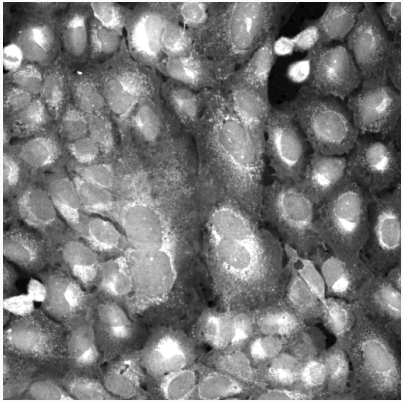
This data as json, CSV (advanced)

| Link | rowid | Compartment | Feature | Channel | Suffix | Feature significance | Cohen's d | \|Cohen's d\| | Perturbation |
|---|---|---|---|---|---|---|---|---|---|
| **243374** | 243374 | Cytoplasm | Texture_Correlation | Mito | _3_00_256 | 0.0 | 3.56 | 3.56 | OPA1 |

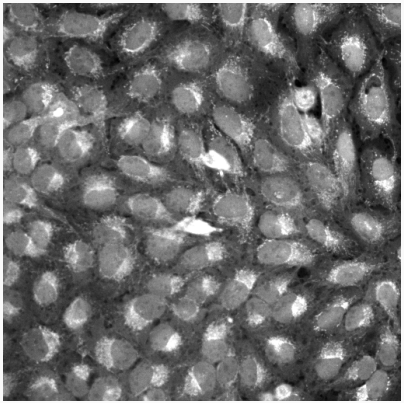
**How:** keep OPA1 and the relevant objects, then export the rows with the reagent ID, filters and source URL.

# JUMP example: OPA1 knockout

**OPA1 CRISPR knockout**



**No-guide negative control**

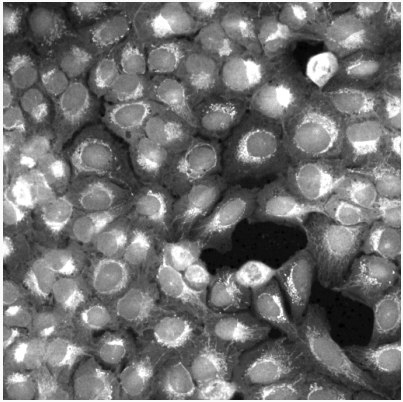


JUMP Mito channel; source_13, CP-CC9-R1-06, D20 / B02, field index 0.

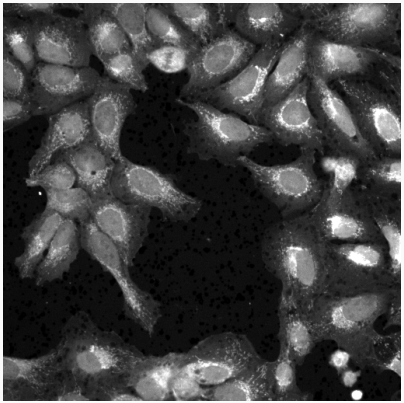
Both panels share intensity limits 219–3985. JUMP data.

# JUMP example: DNM1L knockout

## DNM1L CRISPR knockout



## No-guide negative control



JUMP Mito channel; source_13, CP-CC9-R1-02, P18 / B02, field index 0.

Both panels share intensity limits 187–4463. JUMP data.

# Agents can help to get to the right data quickly

**Start with a biological question**

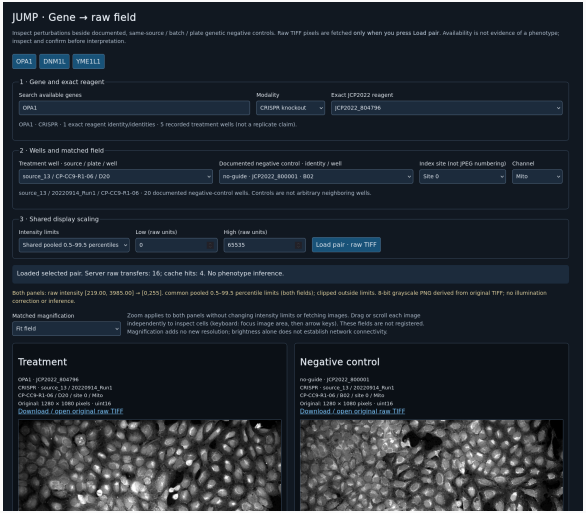
How does disrupting mitochondrial fusion or division change cell images? Find primary studies for OPA1 and DNM1L and explain the expected phenotypes.

**Build a tool to inspect the data:**

“Let me choose a gene and CRISPR reagent, load its raw images beside a plate-matched negative control, and zoom into cells using shared intensity limits.”

**Output:** Interactive viewer.

# AI-assisted tools: gene and raw-image explorer

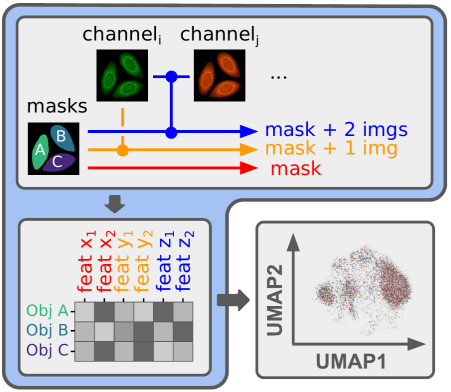



Explorer images: JUMP.

## From a gene to its images

- Select a gene and perturbation.
- Load raw images with plate-matched controls.
- Zoom into cells with shared intensity limits.
- Record which differences are visible.

# Not just JUMP: Extract morphological profiles from your own images




**cp_measure: feature extraction in Python**

- Thousands of features across channels and compartments.
- Fits your Python pipeline.
- Fast software to obtain interpretable morphological features (∼5x CellProfiler).

Adapted from Muñoz et al. (2025), Fig. 1

# Modern tools make image analysis more accessible

| | |
|---|---|
| **Find cells, nuclei and spots** | Cellpose (cells and nuclei); Spotiflow (spots) |
| **Follow cells and organelles over time** | Trackastra / Ultrack |
| **Measure cell and organelle appearance** | cp_measure |
| **Explore your results** | marimo / custom app |

# Applications of JUMP and our tools across biological questions

### Evaluate new lysosomal fluorescence probes

Kenneth Li, University of Edinburgh

- Lysosomal acidification probes in cardiomyocytes.
- Uptake, localization and morphology over time.

### Compare imaging modalities

Minh Doan, GSK Bioimaging Analytics

- Bright-field time series, Cell Painting, cell cycle and viability.
- Which dyes/imaging methods are better for an different Mechanisms of Action

### MisLocus: protein mislocalisation

Johan Fredin, Carpenter-Shen Lab, Purdue University

- Which rare genetic mutations change protein location?
- Compare microscopy of reference and variant proteins.

### Integrated in spatial omics software

Tim Treis, Theis Lab, Helmholtz-Munich

- Integration of morphological features with Squidpy.

# Take-home messages

- JUMP provides public Cell Painting data for around 136,000 chemical and genetic perturbations.
- Biologists can use JUMP Hub to find tested genes and compounds, compare phenotypes and inspect images.
- Open-source image-analysis tools extend morphological profiling to your own experiments.
- Guides and notebooks help scientists or AI agents build analysis workflows.

# Thank you

**Carpenter-Singh Lab**
Anne E. Carpenter
Shantanu Singh

**Cimini Lab**
Beth Cimini
Paula Llanos

**Carpenter-Shen Lab**
Anne E. Carpenter
Runxi Shen

JUMP partners

JUMP supporting partners

JUMP collaborators

Scientific:
- Horizon Discovery, a PerkinElmer Company

Administrative:
- Pistoia Alliance

Academic:
- Umeå University
- Stanford Machine Learning Group

Work on cp_measure and JUMP-lite was supported by GSK and NIH MIRA (R35 GM122547).

# References

**All named tools and models below are clickable.**

**Assay, data and resources**

- Cell Painting (Bray, 2016); profiling review (2021).
- JUMP consortium; counts; centers; images.
- JUMP-lite paper; data/code.
- JUMP Hub; how-to guides; notebooks.
- Guide to Cell Painting.
- Demo recording; Target-2 artifacts.
- word2vec (Mikolov, 2013).

**Image-analysis toolbox**

- cp_measure; paper.
- CellProfiler; paper.
- Cellpose; StarDist.
- Spotiflow: spot detection.
- Trackastra; Ultrack.
- Nahual: model interfaces.
- marimo; uv.
- Squidpy; Morpheus.

# JUMP Hub and slides

JUMP Hub

broad.io/jump

Slides (PDF)

# Morphological profiling: How to turn images into profiles

Start with a simple question: are control and perturbed cells different?

**Classical features (e.g., CellProfiler)**

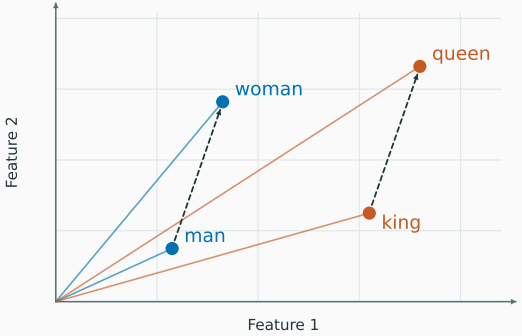
- Size, shape, intensity, texture and relationships between channels and objects.
- Interpretable but partially redundant.

**Learned representations**

- Features learned from images by neural networks (deep learning).
- Non-redundant numerical vectors (embeddings).

# Words as vectors: word2vec

Learning can be viewed as compression




Word → vector

Learned from context

man → woman
king → queen

$$v(\text{king}) - v(\text{man}) + v(\text{woman}) \approx v(\text{queen})$$

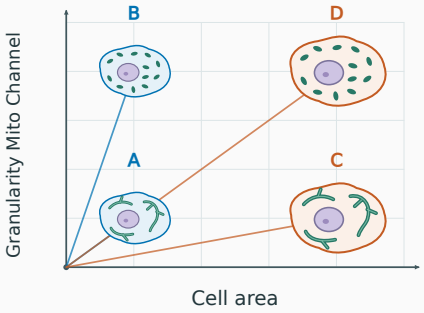
Mikolov et al. (2013)

# From words to cells: vectors for comparison

## Classical features

Defined measurements

$x = [\text{cell area}, \text{granularity}, \ldots]$

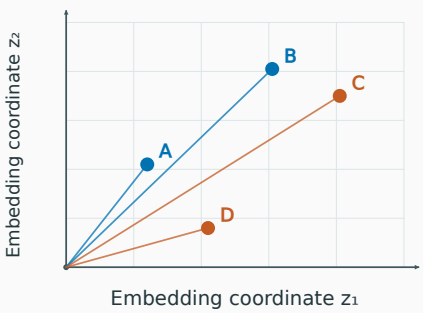



## Learned embeddings

Learned from images

**Not necessarily directly interpretable.**

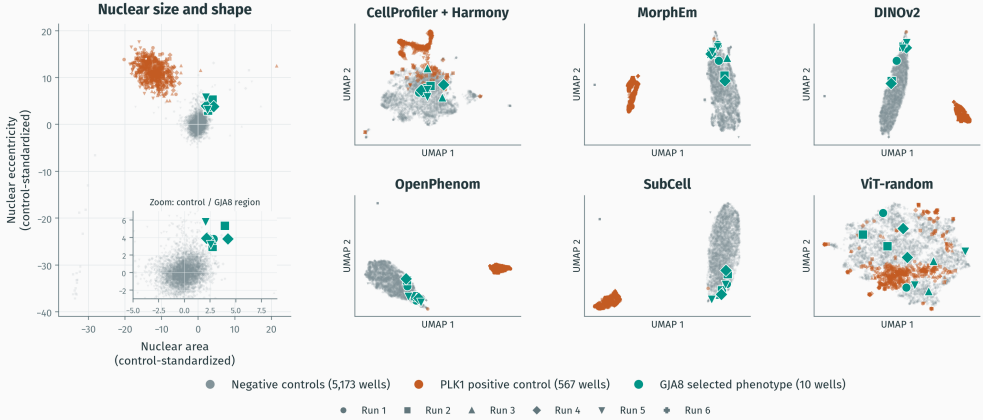
$z = [z_1, z_2, \ldots, z_n]$




Control

Perturbation

# Positive controls contextualize perturbations in both learned and classical representations




Data: JUMP-lite (2026); models: MorphEm, DINOv2, OpenPhenom, SubCell; CellProfiler/Harmony recipe

# JUMP-lite: a compact benchmark

Same images. Different methods.

Test whether profiles recover known relationships.

Same selected images: 8.7 TB $\rightarrow$ 92 GB ($\sim$94$\times$ smaller; lossy JPEG XL).

Adapted from Muñoz et al. (2026)

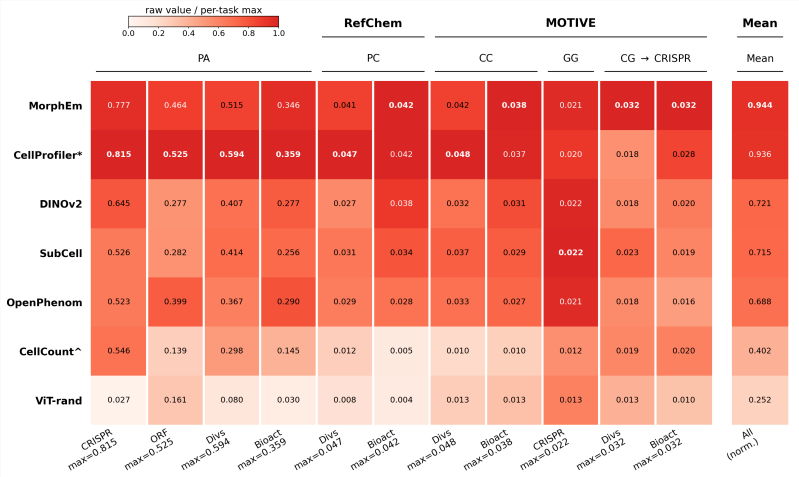
# Nahual: swap models within one workflow

## Swap models. Keep your workflow.

| | |
|---|---|
| **Image features** | DINOv2 ↔ DINOv3 ↔ OpenPhenom ↔ MorphEm |
| **Segmentation** | Cellpose ↔ StarDist |
| **Spot detection** | Spotiflow |
| **Tracking** | Trackastra ↔ Ultrack |

# JUMP-lite: comparing image representations

Recover replicate perturbations and known biological relationships.

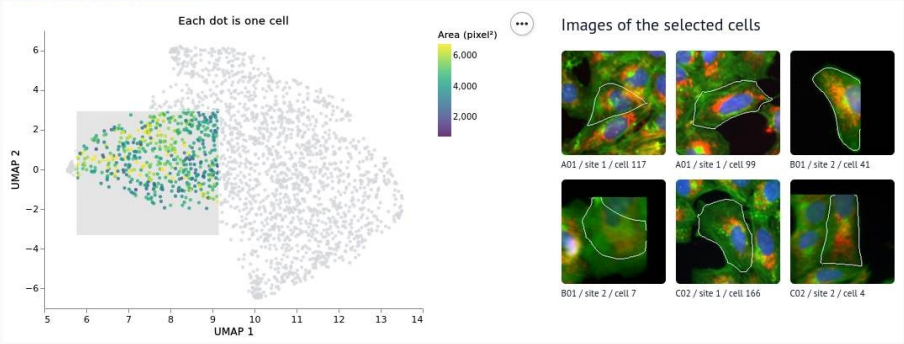



Raw images; colors normalized within each task. CellProfiler/count references use more sites per well.

Adapted from Muñoz et al. (2026), Figure 5

# Backup: explore cell subpopulations

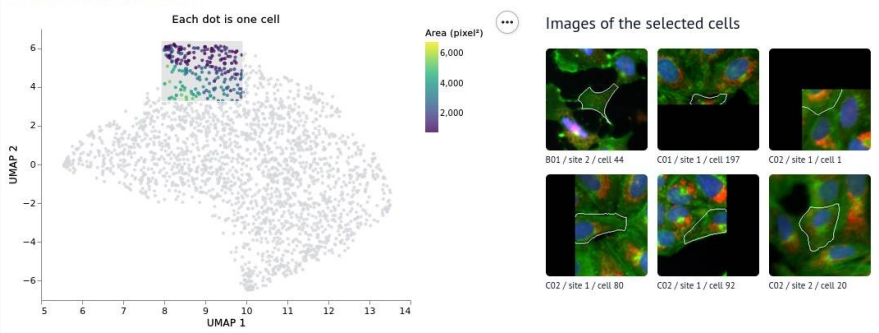
Interactive marimo notebook — separate recorded example




Each dot is a cell: 2,223 from one Target-2 plate. Shown selection: 545 cells. Inspect images, area and other features.

Source: Muñoz (2026), JUMP Target-2 notebook recording

# Backup: compare another cell selection




- Second selection: 224 cells; compare images and measurements.
- Not the compound queries; no treatment/control assignments in this example.

Source: Muñoz (2026), JUMP Target-2 notebook recording